Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

MAT 2306 – Operational Research I
Tutorial No: 3

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1. Solve the following linear programming problem by using the two-phase simplex method

$$MinZ = x_1 - 2x_2 - 3x_3$$
$$s.t{:}\ -2x_1 + x_2 + 3x_3 = 2$$
$$2x_1 + 3x_2 + 4x_3 = 1$$
$$x_1, x_2, x_3 \geq 0$$

2. Solve the following linear programming problem by using the two-phase simplex method

$$MinZ = x_1 + x_2$$
$$s.t{:}\ 2x_1 + x_2 \geq 4$$
$$x_1 + 7x_2 \geq 7$$
$$x_1, x_2 \geq 0$$

3. Solve the following transportation problem by using the Northwest Corner Method

	D_1	D_2	D_3	D_4	***Supply***
S_1	19	30	50	10	**7**
S_2	70	30	40	60	**9**
S_3	40	8	70	20	**18**
Demand	**5**	**8**	**7**	14	

4. Solve the following transportation problem by using the Northwest Corner Method and check the solution's optimality using the Modified Distribution Method

	X	***Y***	***Z***	***Availability***
A	4	3	2	**10**
B	5	6	1	**8**
C	6	4	3	**5**
D	3	5	4	**6**
Requirement	**7**	**12**	**5**	

5. Determine an initial basic feasible solution to the following transportation problem using
   I. North West Corner method
   II. Least Cost Method

	D_1	D_2	D_3	D_4	***Supply***
S_1	6	4	1	5	**14**
S_2	8	9	2	7	**16**
S_3	4	3	6	2	**5**
Required	**6**	**10**	**15**	**4**	

6. The company has decided to initiate the production of some or all of four new products at 3 branches plants with excess production capacity. The available capacities of in the three plants are 40, 50 and 30 units per week while the demand for the four products are 30, 30, 40, 20 units respectively. The cost per unit of production in the plants is given in following table.

	D_1	D_2	D_3	D_4	***Supply***
S_1	3	6	5	2	**40**
S_2	4	9	5	4	**50**
S_3	3	4	4	3	**30**
Required	**30**	**30**	**40**	**20**	

Find the IBFS using VAM and obtain the quantities of production at the optimum by Modified Distribution Method.

(Please answer the all the question and submit the answer script to the tutorial room on or before 17th of December 2024)